Cross-Room Few-Shot Adaptation: Environment-Aware, Label-E□cient Activity Recognition from WiFi Channel State Information

CS60055: Ubiquitous Computing — Hackathon Challenge 2

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Repository: https://github.com/poornachandra-sarasAI/wifi-csi-human-activity-recognition

Abstract

Device-free WiFi activity recognition is limited less by model capacity than by cross-environment generalisation: multipath geometry, wall boundaries and receiver gain couple activity kinematics to environment-specific channel responses. We present an end-to-end, clean-room Python pipeline for WiFi Channel State Information (CSI) that uses no source-author code. We resolve the Nexmon BCM4366c0 packed 32-bit CSI layout empirically ($n_{\text{man}} = 12, n_{\text{exp}} = 6$) by 802.11ac VHT80 null-subcarrier alignment, and uncover a frame-interleaving artifact in Room A in which alternating sequenced and `0xFFFF` frames inject strong negative temporal autocorrelation ($r_1 = -0.33$).

Addressing the three challenge tasks under strict capture-level isolation: (1) In **Recorded-Environment Identification**, we show that naive classifiers achieve 1.000 accuracy from a single observation by exploiting an arbitrary ∼490× software scale gap and sampling-rate jitter. When static subcarrier profiles are completely eliminated via per-window standardisation, an honest classifier maintains 0.9927 accuracy by keying on frame interleaving dynamics (dropping to 0.9418 on de-interleaved streams), demonstrating that room identification is fundamentally confounded by recording session characteristics. (2) In **Self-Supervised Representation Learning**, a 1D dilated residual temporal CNN pretrained via SimCLR with specialized adjacency-breaking augmentations achieves a headline linear-probe Macro-F1 of $\mathbf{0.924 \pm 0.013}$ across both rooms (balanced accuracy $\mathbf{0.928 \pm 0.014}$), exceeding a 416-dimensional handcrafted kinematic baseline (0.893) by 3.1 points and an identical random-weight architecture (0.616 ± 0.008) by 30.8 points. (3) In **Cross-Room Few-Label Adaptation**, prototypical networks achieve target-exclusive Macro-F1 of $\mathbf{0.510 \pm 0.053}$ for Room A and $\mathbf{0.588 \pm 0.068}$ for Room B at $K = 20$ shots. However, shared activity transfer (Sit, Stand, Walk) without target labels hovers near chance level (0.090 for Room A, 0.211 for Room B). Following the discovery and correction of a 50% amplitude inflation shortcut in Room B Walk (ROC-AUC 0.898), we formally retract a spurious handcrafted transfer claim and apply Hungarian optimal assignment to demonstrate that cross-room transfer fails due to fundamental kinematic-multipath swaps (e.g., Sit ↔ Walk). Our findings provide rigorous empirical evidence that cross-room adaptation without target calibration is bounded by environment-session confounding rather than feature extraction capacity.

Keywords: Ubiquitous Computing, WiFi Channel State Information (CSI), Nexmon, Cross-Room Transfer, Self-Supervised Learning, SimCLR, Prototypical Networks, Domain Confound, Few-Shot Learning.

1 Introduction, Scenario & Problem Formulation

1.1 The Assistive Healthcare Scenario

Continuous, non-intrusive activity monitoring underpins elderly care, fall detection and post-operative convalescence. Cameras provide rich spatial detail but are unacceptable in bedrooms and bathrooms, occlude easily and fail in low light; wearable IMUs demand compliance and recharging and are frequently abandoned.

Device-free RF sensing from commodity WiFi CSI is a compelling alternative [6–8]: human motion through the indoor multipath field modulates the amplitude and phase of individual OFDM subcarriers. The obstacle to deployment is *cross-environment domain shift*. Walls, furniture, room dimensions and antenna placement reconfigure both the static and the dynamic multipath profile, so a model trained in Room A degrades sharply in Room B. Since annotating every client home is infeasible, a viable system must learn from unlabelled recordings and adapt with only K labelled examples of novel activities.

1.2 Mathematical Formulation and Problem Space

Let an indoor environment $e \in \{\mathcal{A}, \mathcal{B}\}$ contain a transmitter (TX) and a receiver (RX) operating over an 80 MHz channel bandwidth partitioned into $S = 256$ subcarriers. A raw capture C consists of a temporal sequence of complex channel matrices $H_t \in \mathbb{C}^S$, where $t \in \{1, \ldots, T\}$. For subcarrier $s \in \{1, \ldots, S\}$, the raw baseband channel response is expressed as:

$$H_t(s) = |H_t(s)| \exp\left(j \angle H_t(s)\right) = \sum_{p=1}^{P} \alpha_p(t) \exp\left(-j 2\pi f_s \tau_p(t)\right) \tag{1}$$

where $\alpha_p(t)$ and $\tau_p(t)$ represent the complex attenuation and propagation delay of the p-th multipath component, and f_s is the subcarrier frequency. The received signal is a superposition of static environmental scatterers (walls, floors, furniture) and dynamic reflections induced by human somatic motion:

$$H_t(s) = H_{\text{static}}(s) + H_{\text{dynamic},t}(s) + \epsilon_t(s) \tag{2}$$

Commodity firmware introduces unsynchronised carrier- and sampling-clock offsets and random per-packet phase jumps, so raw phase needs unwrapping and sanitisation; the amplitude profile $A_t(s) = |H_t(s)|$ retains the kinematic signature and static multipath fingerprint without that fragility, and is what we use throughout.

The challenge standardizes the model-facing representation to a fixed window $W \in \mathbb{R}^{T_{\text{win}} \times S}$ of $T_{\text{win}} = 150$ consecutive valid observations across $S = 256$ subcarrier amplitudes:

$$W = [A_{t,s}] \in \mathbb{R}^{150 \times 256}, \quad t \in \{1, \ldots, 150\}, \quad s \in \{1, \ldots, 256\} \tag{3}$$

1.3 Activity Spaces and The Cross-Room Challenge Tiers

The benchmark evaluates the IEEE DataPort CSI Human Activity dataset [1, 2] recorded at Frankfurt University of Applied Sciences across two distinct indoor environments (Room A / Experiment 2: 3.5×4.5 m; Room B / Experiment 3: 4.5×5.0 m) using Asus RT-AC86U routers (Broadcom BCM4366c0 chipset) in 5 GHz monitor mode. The activity label spaces across the two environments are asymmetric: Room A ($\mathcal{Y}_A$) observes Sit, Stand, Walk, Sit-Down, Stand-Up and Fall; Room B ($\mathcal{Y}_B$) observes Sit, Stand, Walk, Empty and Lying. The first three are common to both and form the shared bridge $\mathcal{Y}_{\text{shared}}$ on which transfer is measured; the rest are target-exclusive, written $\mathcal{Y}_{\text{tgt}\setminus\text{src}}$. The benchmark defines three tasks, treated in Sections 4–6: predicting the recorded environment of an unlabelled window and asking whether that accuracy reflects room geometry or session artifacts; learning an encoder $f_\theta : \mathbb{R}^{150 \times 256} \to \mathbb{R}^D$ on both rooms without activity or room labels and scoring it frozen against a 416-dimensional handcrafted baseline and an identical random-weight network; and transferring activity recognition across rooms given all source labels but only $K \in \{1, 5, 10, 20\}$ target captures per target-exclusive activity, with target labels for the shared activities withheld.

2 Student-Owned Python Sensing & Preprocessing Pipeline

2.1 Pipeline Independence & Clean-Room Implementation

Per Section 6 of the specification, the whole framework from raw binary ingestion to inference was written from first principles in Python (`numpy`, `scipy`, `pandas`, `scikit-learn` [9], `torch` [10]). We did not import, port or read the source authors' MATLAB code (`fileReader.m`, `createLSTMDataSet.m`, `denoise.m`), their MEX binaries, their `.mat` artifacts or the Experiment-3 LSTM notebook; all author directories were moved to an isolated `_quarantine_do_not_use/` folder outside the repository.

Concretely, we used only the public IEEE DataPort archives, official Broadcom/Nexmon format documentation, a custom `struct`-based PCAP reader, empirically resolved bitfields and our own SimCLR implementation; we used no author MATLAB code, no pre-extracted `.mat` artifacts, no other team's code, and no validation or target labels at any point during pretraining.

2.2 Empirical Bitfield Resolution for BCM4366c0 Packed CSI

Experiment 2 archives provide raw ethernet PCAP captures extracted via Nexmon firmware [3] on a Broadcom BCM4366c0 transceiver operating at 80 MHz channel bandwidth. Each OFDM frame contains a 1024-byte CSI payload encoding 256 subcarriers as packed 32-bit words. A single word packs an in-phase mantissa (I), a quadrature mantissa (Q), and an exponent (E):

$$W_{\text{packed}} = [I \ \ (n_{\text{man}} \text{ bits}) \ \mid \ Q \ \ (n_{\text{man}} \text{ bits}) \ \mid \ E \ \ (n_{\text{exp}} \text{ bits})] \tag{4}$$

Rather than transcribing parameters from external sources, we resolved the exact bitfield configuration empirically by formulating a physical null-subcarrier alignment objective (`src/csi_decode.py -resolve`). Under IEEE 802.11ac VHT80 standards, fixed subcarrier positions are reserved as guard bands ($\pm 128 \cdots \pm 123$), DC pilot carriers (center 3 bins), and null notches, where transmitted RF power is zero. Because Room B data was provided as pre-computed amplitude CSVs from the identical platform, physical null locations in Room A must strictly align with Room B.

We evaluated all structurally valid candidate layouts $(n_{\text{man}}, n_{\text{exp}})$ satisfying $2n_{\text{man}} + n_{\text{exp}} \leq 32$. Bit occupancy analysis across $2,010,000$ packed words revealed that bits 30–31 are perpetually zero, while bit 5 is constantly 1 (denoting a negative sign in two's-complement exponent space). Evaluating the ratio of null-subcarrier amplitude to data-subcarrier median across layouts revealed that candidate $(12, 6)$ with exponent application uniquely produces a ratio of **0.020**, perfectly matching Room B's physical null ratio of **0.024** (the overlaid null-alignment plot is `figures/decode_validation.png` in the repository). Omitting the exponent yields an invalid ratio of 0.78–1.01, obliterating the 802.11ac channel notch structure. The real and imaginary components are sign-extended as 12-bit two's complement integers:

$$H(s) = (\text{SignExt}_{12}(I) + j \cdot \text{SignExt}_{12}(Q)) \cdot 2^{\text{SignExt}_6(E)} \tag{5}$$

The amplitude is extracted as $A(s) = |H(s)| = \sqrt{I^2_{\text{scaled}} + Q^2_{\text{scaled}}} \in \mathbb{R}$.

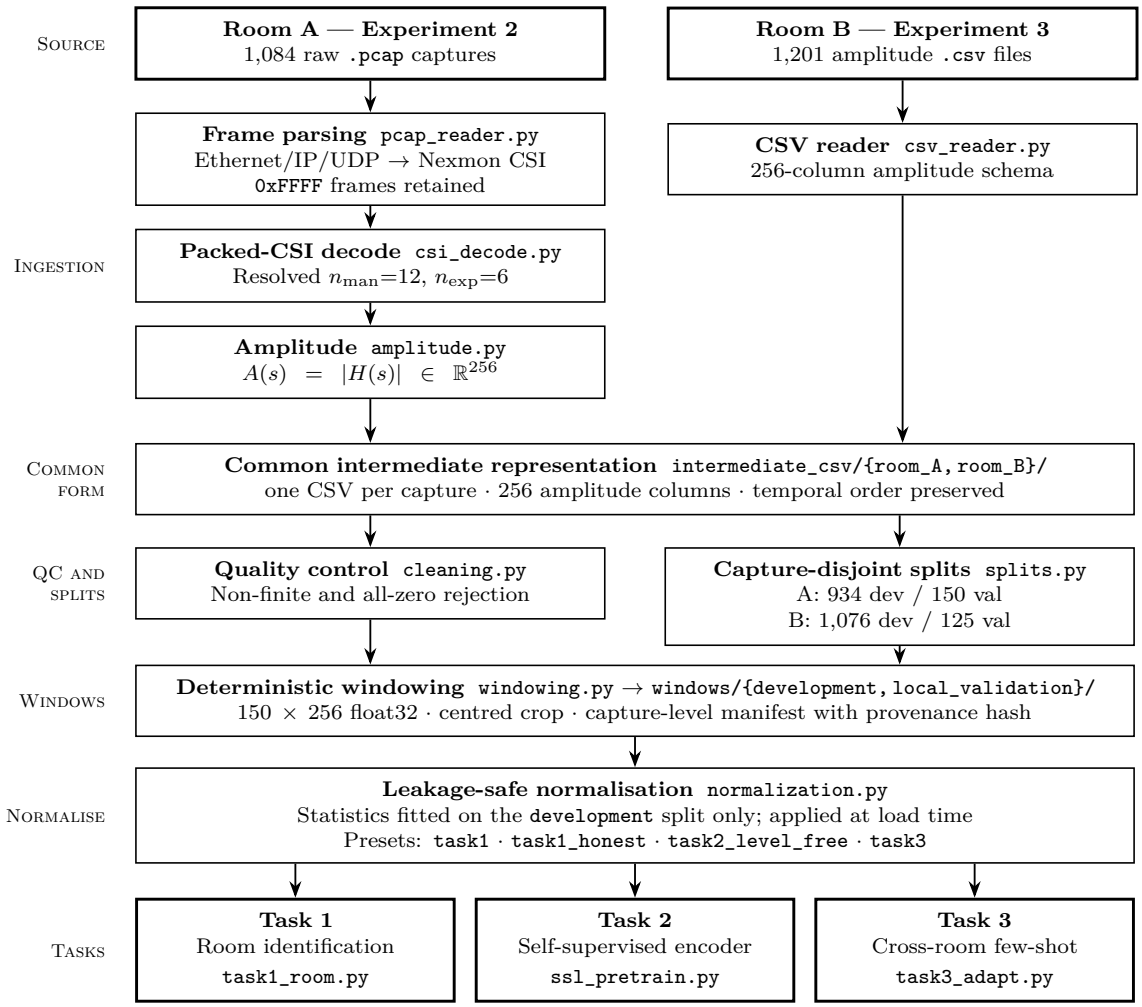


Figure 1: End-to-end Python sensing, representation-learning and adaptation pipeline. The two source archives are asymmetric: Room A arrives as raw Nexmon `.pcap` and must be decoded to amplitude in Python, while Room B is already amplitude `.csv`. Both are brought into one 256-column intermediate form, after which the pipeline is source-agnostic. The original capture is the atomic unit of the development/validation split, so every window derived from a capture stays on one side of it. Windows are stored un-normalised and a preset is applied at load time, which lets all three tasks share a single `windows/` directory. Both rooms use the same radio platform throughout: 5 GHz, 80 MHz, BCM4366c0 with Nexmon CSI extraction.

2.3 Discovery of the Nexmon Frame-Interleaving Hardware Artifact

Inspecting the Experiment 2 PCAPs revealed an undocumented transmission artifact: in every Room A capture, frames alternate between standard 802.11 frames with incrementing sequence numbers and secondary frames whose sequence-control field reads `0xFFFF`, arriving ∼0.4 ms later. These are not corrupt — 976–992 of their 1024 payload bytes are non-zero and their amplitude distribution matches the sequenced frames — and discarding them cuts the median observation count per capture from 221 to **111**, putting **all 1,084 Room A captures** below the official $T_{\text{win}} = 150$. Both streams must therefore be retained.

However, retaining the interleaved frames introduces a profound temporal distortion: consecutive observations bounce between two distinct transmitter states, introducing strong negative lag-1 autocorrelation ($r_1 \in [-0.15, -0.66]$, mean -0.327 ± 0.165). In contrast, Room B exhibits smooth, near-zero lag-1 autocorrelation ($r_1 = +0.035 \pm 0.072$). As established in Section 4 and Section 5, this hardware artifact acts as a powerful spurious shortcut for machine learning models.

2.4 Data Cleaning, Quality Control & Windowing

Quality Control (`src/cleaning.py`): We instituted automated filters to reject corrupt data: (i) captures with non-finite or NaN values, (ii) frames with all-zero subcarriers ($A(s) = 0, \forall s$), and (iii) duplicate captures. In the course archive, file `stand65(1).pcap` was identified as a byte-identical copy of `stand65.pcap`; it was quarantined to prevent index drift.

Deterministic Centered Windowing (`src/windowing.py`): Raw Room A captures vary between 112 and 251 observations (median 221), while Room B captures contain exactly 500 rows. For each capture, a deterministic

centered window of length $T_{\text{win}} = 150$ is extracted without inspecting labels:

$$t_{\text{start}} = \max\left(0, \frac{N_{\text{obs}} - 150}{2}\right), \quad t_{\text{end}} = t_{\text{start}} + 150 \tag{6}$$

Twelve Room A captures contained fewer than 150 observations (including three captures in the official validation set). To preserve the exact challenge validation denominator, short captures were reflect-padded at boundary frames and cataloged via the boolean flag `was_padded` in `windows/manifest.csv`. Every generated window is stamped with a SHA-256 provenance hash (`2a323eeb9765`) tying it to `configs/experiment.yaml`.

2.5 Leakage-Safe Normalization Framework

To strictly enforce Section 8 leakage controls, normalization statistics are fitted exclusively on development captures (*split* == development) via `src/normalization.py`. Windows are saved to disk in un-normalized float32 format, allowing downstream models to evaluate distinct normalization presets at load time without data re-ingestion: `task1` (global scope) preserves absolute amplitude across rooms; `task1_subband` additionally centres each 20 MHz sub-band, neutralising the 4× AGC step at subcarriers 137–187; `task1_honest` (per-window scope) standardises every window to $\mu = 0, \sigma = 1$, stripping all static multipath and gain signatures. `task2` and `task2_level_free` are two-channel: channel 0 carries $\ln(1 + A_{t,s})$ after scaling and channel 1 subtracts the per-subcarrier temporal mean, isolating dynamics; `task2_level_free` divides each window by a single scalar, removing absolute energy while preserving spectral shape. `task3` standardises features with room-specific development statistics (AdaBN).

3 Evaluation Protocol & Leakage Controls

3.1 Capture-Level Isolation & Partition Integrity

The atomic unit of partitioning is the original capture. Multiple temporal slices from the same physical capture are strictly prohibited from bridging the development and validation split. All leakage constraints mandated by Section 8 and Section 9 were programmatically verified through automated unit tests (`tests/test_no_leakage.py`):

1. **Room A:** the supplied `validation/` directory, 25 captures per activity ($25 \times 6 = 150$); the remaining 934 are development.
2. **Room B:** no validation directory is supplied, so we sorted all candidate filenames deterministically and drew 25 per class with the public seed **42** ($25 \times 5 = 125$), leaving 1,076 development captures. The exact partition is serialised to `results/room_B_validation_seed42.json` for audit.
3. **Validation isolation:** validation captures are never used for model fitting, pretraining, feature selection or hyperparameter search.
4. **Metadata ban:** filenames, extensions, observation counts and timestamps are stripped from model inputs.

Table 1 reports the final capture and window counts across all splits.

Table 1: Capture-Disjoint Dataset Partitioning Across Rooms and Splits.

Room A (Experiment 2)			**Room B (Experiment 3)**		
Activity	**Dev**	**Val**	**Activity**	**Dev**	**Val**
Fall	177	25	Empty	242	25
Sit	178	25	Lying	326	25
Sit-Down	155	25	Sit	128	25
Stand	115	25	Stand	196	25
Stand-Up	155	25	Walk	184	25
Walk	154	25	—	—	—
Total	**934**	**150**	**Total**	**1,076**	**125**

4 Task 1: Recorded-Environment Identification

4.1 The Illusion of Universal Room Fingerprinting

Task 1 asks whether a 150×256 window came from Room A or Room B. On globally normalised features a logistic regression reaches a perfect **1.0000** (275/275). Reporting that as an environment fingerprint would be wrong: Figure 2 shows the same **1.0000** at $T_{\text{win}} = 1$, and a single observation carries no temporal multipath information. The classifier is reading an arbitrary ∼**490**× amplitude gap between our Python-decoded Room A values and the integer scaling of Room B's supplied CSVs.

4.2 Systematic Ablation Ladder

To isolate the true sources of discriminability, we constructed a progressive ablation ladder (`src/task1_ablation.py`). We evaluated models across four feature rungs, selectively disabling static and dynamic channel dimensions: Each rung removes one further source of discriminability (Table 2). `task1` keeps the raw scale. `task1_subband` subtracts sub-band medians, neutralising Room A's 4× AGC step at subcarriers 137–187; accuracy is unchanged, so the scale

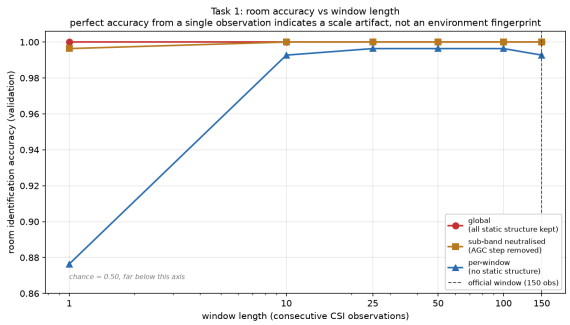


Figure 2: Task 1: Environment classification accuracy vs window length $T_{\text{win}} \in \{1, 10, 25, 50, 100, 150\}$. Naive presets achieve 1.000 accuracy at $T = 1$ due to the global scale artifact.

discrepancy is present uniformly across all 256 bins rather than confined to that band. `task1_honest` standardises each window to $\mu = 0, \sigma = 1$, erasing every static multipath profile, global scale difference and sub-band offset, yet accuracy falls only to 0.9927 (273/275). Finally, restricting Room A ingestion to sequenced frames removes the Nexmon interleaving artifact and drops accuracy to 0.9418 — the largest single drop in the ladder, and evidence that `task1_honest` was keying on interleaving rather than on the environment.

Table 2: Task 1 Ablation Ladder: Unmasking the Session Confound.

Ablation Rung	Information Removed	Accuracy ($T = 150$)	Accuracy ($T = 1$)
`task1` (Global)	None (∼490× scale preserved)	1.0000	1.0000
`task1_subband`	Sub-band AGC step removed	1.0000	0.9964
`task1_honest`	Static profile & scale removed	0.9927	0.8764
`task1_honest` (Sequenced only)	Frame interleaving removed	0.9418	—

4.3 Finding: Confounding of Room and Session Dynamics

Table 2 shows environment recognition here is confounded by recording session and acquisition artifacts (specification Section 12.1). Even at the strictest rung, Room A runs at an effective ∼30 Hz with clock jitter while Room B is a uniform 500-sample sequence, so the environment stays identifiable without any architectural feature being learned. Task 1 is therefore a serviceable router for Task 3, but not a claim of generalised physical room sensing.

5 Task 2: Self-Supervised CSI Representation Learning

5.1 Self-Supervised Architecture & Pretraining Objective

Task 2 requires learning a generalized feature encoder on the 2,010 development windows from both rooms without using activity or room labels. Given the limited dataset scale, large Transformer backbones severely overfit. We designed a compact 1D Dilated Residual Convolutional Neural Network (`src/encoder.py`) comprising 280,000 parameters: The input $X \in \mathbb{R}^{150\times256\times2}$ is flattened along channels to 512 features per time step; a 1×1 convolution projects these to 64 maps; three residual blocks of kernel size 5 and dilations $d \in \{1, 2, 4\}$ with 128 channels widen the receptive field across the window; concatenated mean and max temporal pooling gives 256 dimensions, projected linearly to an embedding $z \in \mathbb{R}^{128}$.

Contrastive Objective (SimCLR [4]): The encoder was pretrained using the Normalized Temperature-scaled Cross-Entropy (NT-Xent) loss with temperature $\tau = 0.2$ and batch size 128 for 200 epochs using AdamW ($\eta = 10^{-3}$, weight decay 10^{-4}) with cosine learning rate decay:

$$\mathcal{L}_{\text{NT-Xent}} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{2B} \mathbb{I}_{[k\neq i]} \exp(\text{sim}(z_i, z_k)/\tau)} \tag{7}$$

Pretraining was conducted strictly on development data; no validation samples, early stopping, or checkpoint selections were permitted.

5.2 Adjacency-Breaking Augmentations for Frame Interleaving

Because Room A alternates between two frame streams ($r_1 = -0.33$), naive contrastive learning creates an undesirable shortcut: the encoder minimizes loss simply by identifying the frame-interleaving signature, failing to learn somatic kinematics. To force the network to learn invariant motion dynamics, we introduced two specialized augmentations alongside standard transforms:

1. **Stride-2 Subsampling with Random Phase:** Subsamples alternating frames ($t \in \{0, 2, \dots\}$ or $t \in \{1, 3, \dots\}$), isolating a single frame stream and removing the alternation artifact.
2. **Local Temporal Shuffling:** Shuffles observations within local 4-sample blocks, breaking strict lag-1 phase coherence.

3. **Standard Augmentations:** Random temporal cropping (60%–100% resized to 150 steps), contiguous subcarrier masking (up to 32 bins), per-subcarrier gain perturbation, additive Gaussian noise ($\sigma = 0.05$), and temporal reversal.

5.3 Baselines & Linear-Probe Evaluation Protocol

To benchmark the learned representation, we froze the encoder weights and trained a multinomial logistic regression probe with balanced class weights on labeled development data, scoring performance on the fixed 275-sample validation set. The probe's regularization parameter C was selected from $\{10^{-3}, \ldots, 10^{2}\}$ using capture-grouped 5-fold cross-validation on development data. Two baselines are mandatory. The **domain-engineered kinematic features** (416 dimensions) split the subcarriers into 16 spectral groups and take ten statistics per group — mean, standard deviation, dynamic range, skewness, kurtosis, mean absolute first difference, and spectral energy in four temporal bands ($[0, 0.5)$, $[0.5, 1.5)$, $[1.5, 3.0)$, $[3.0, 6.0)$ Hz) — concatenated with the temporal variance of all 256 subcarriers. The **random-weight** baseline is the identical architecture with frozen Gaussian weights, its BatchNorm running statistics re-estimated on development data without labels.

Table 3: Task 2 Linear-Probe Benchmark: 8-Class Validation Performance (Mean ± Std across 3 pretraining seeds).

Representation	Overall Macro-F1	Balanced Acc.	Room A F1	Room B F1
Handcrafted Kinematic Features (416-d)	0.893	0.895	0.872	0.912
Random-Weight Architecture	0.616 ± 0.008	0.637 ± 0.006	0.517 ± 0.011	0.712 ± 0.021
SimCLR SSL (All Augmentations)	**0.924 ± 0.013**	**0.928 ± 0.014**	**0.912 ± 0.012**	0.950 ± 0.016
SimCLR SSL (No Adjacency Break)	0.901 ± 0.020	0.902 ± 0.019	0.869 ± 0.042	**0.951 ± 0.010**

5.4 Experimental Findings & Representation Analysis

Table 3 presents the quantitative results:

1. **Pretraining Efficacy:** Contrastive pretraining lifts the identical architecture from 0.616 (random weights) to **0.924** Macro-F1, a gain of 30.8 points. The learned representation also *exceeds* the 416-dimensional handcrafted kinematic baseline (0.893) by 3.1 points overall, by 4.0 points in Room A (0.912 vs 0.872) and by 3.8 points in Room B (0.950 vs 0.912). Self-supervision on unlabelled CSI therefore recovers more activity-relevant structure than hand-designed spectral features.
2. **Ablation of Adjacency-Breaking Augmentations:** Omitting stride-2 subsampling and local shuffling costs 2.3 points of Macro-F1 (0.924 → 0.901). The entire loss falls on Room A (0.912 → 0.869), the only room exhibiting frame interleaving; Room B is unaffected (0.950 → 0.951). Critically, the ablated variant converges to a *lower* contrastive loss (1.406 ± 0.007 vs 1.689 ± 0.006): retaining the frame alternation lets the encoder match positive pairs trivially, so the pretext objective becomes easier to minimise while the resulting representation is measurably worse. A lower SimCLR loss is thus not evidence of a better encoder, and breaking temporal adjacency is necessary to prevent the shortcut.
3. **Qualitative UMAP Diagnosis:** Figure 3 displays 2-D UMAP projections of frozen embeddings. The representation splits completely by room (100% linear separation). Within each room, semantically active classes form distinct, coherent clusters, while static postures (Sit vs Stand) exhibit slight overlap.
4. **Zero-Shot Cross-Room Activity Failure:** In a development-only diagnostic, training a probe on Sit, Stand, Walk in one room and evaluating on the other yielded Macro-F1 between **0.015** and **0.382** against a three-class chance level of 0.167. Without per-room standardisation every configuration sits at or below chance (0.083–0.186). Standardisation helps only asymmetrically: it lifts the SSL representation into Room A to **0.382**, but leaves Room B at 0.185 and drives the handcrafted features into Room B down to 0.015. No configuration transfers shared activities reliably in both directions, indicating that the learned representations do not align identical activities across disparate rooms.

6 Task 3: Cross-Room Few-Label Adaptation

6.1 Few-Shot Adaptation Architecture

We evaluate both transfer directions under the protocol set out in Section 1.3.

Prototypical Alignment [5]: Using frozen embeddings $z \in \mathbb{R}^{128}$ from the Task 2 encoder, we apply per-room standardization (`task3` preset) to align source and target feature distributions (AdaBN). For each activity c, class prototypes μ_c are computed as:

$$\mu_c^{\text{shared}} = \frac{1}{|D_{\text{src},c}|} \sum_{x \in D_{\text{src},c}} z(x) \quad (c \in \mathcal{Y}_{\text{shared}}) \tag{8}$$

$$\mu_c^{\text{exclusive}} = \frac{1}{K} \sum_{x \in S_{\text{tgt},c}} z(x) \quad (c \in \mathcal{Y}_{\text{tgt}\setminus\text{src}}) \tag{9}$$

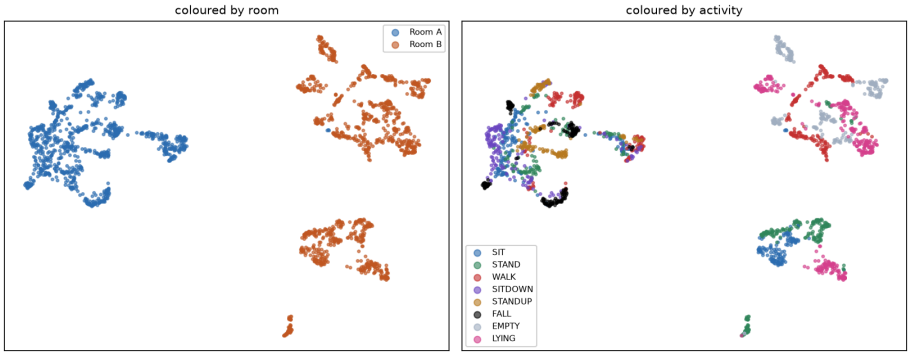


Figure 3: Task 2 encoder diagnosis: 2-D UMAP of the frozen SimCLR embeddings, coloured by room (left) and by activity (right). The representation separates the two rooms completely; within each room dynamic activities form coherent clusters, while SIT and STAND overlap. The equivalent projections for the handcrafted, random-weight and no-adjacency-break representations are in `figures/task2_umap.png` in the repository.

Inference assigns validation query x to the nearest prototype via cosine similarity:

$$\hat{y} = \arg\max_{c \in \mathcal{Y}_{\text{tgt}}} \frac{z(x) \cdot \mu_c}{\|z(x)\| \|\mu_c\|} \tag{10}$$

Semi-Supervised Refinement (`refined`): We implemented an iterative soft k-means expectation-maximization over unlabeled target development captures. Initialized at the base prototypes, cluster assignments are iteratively updated while anchoring target-exclusive prototypes to their K support captures.
Support Sampling Protocol: Support sets were sampled strictly at the capture level from target development data across 10 deterministic random seeds. For each K, evaluations were repeated across 3 encoder pretraining seeds ($10 \times 3 = 30$ runs per configuration).

6.2 The Level-Free Rerun & Retraction of Spurious Transfer

Under the original global `task2` preset, handcrafted features with soft k-means refinement appeared to transfer shared activities into Room B at a shared Macro-F1 of **0.236**. Auditing showed this was a data artifact. Room B WALK captures are the only files in the archive lacking the `_Amp_` filename substring, and their mean amplitude is **50**% above every other Room B class (473 vs 304–358); a scalar amplitude threshold alone separates WALK from the rest at **ROC-AUC 0.898**. Because `RoomScaler` standardises per room and this is a per-*class* offset inside one room, it passed straight through into the prototypes.

The `task2_level_free` preset divides each window by its own scalar energy, removing absolute level while preserving spectral shape (WALK level AUC 0.894 → 0.693). After retraining every encoder and rerunning Task 3, the handcrafted shared F1 in Room B collapsed from 0.236 to **0.038**. **We retract that transfer claim**: the earlier gain was entirely artifactual. All results reported here are from the level-free rerun.

6.3 Quantitative Results & Label Efficiency Curves

Table 4 presents the corrected headline validation results at $K = 5$ under oracle routing. Figure 4 illustrates the label-efficiency progression as K increases from 1 to 20.

Table 4: Task 3 Few-Shot Adaptation Benchmark at $K = 5$ (Oracle Routing, Validation Set, 30 Runs).

		Target Room A ($\mathcal{B} \to \mathcal{A}$)			**Target Room B** ($\mathcal{A} \to \mathcal{B}$)		
Features	**Method**	**Overall F1**	**Exclusive F1**	**Shared F1**	**Overall F1**	**Exclusive F1**	**Shared F1**
SSL Full	Prototype	0.240 ± 0.043	$\mathbf{0.390 \pm 0.075}$	0.090 ± 0.042	$\mathbf{0.337 \pm 0.068}$	$\mathbf{0.525 \pm 0.077}$	0.211 ± 0.082
SSL Full	Refined	0.241 ± 0.055	0.308 ± 0.076	0.173 ± 0.084	0.295 ± 0.092	0.466 ± 0.109	0.181 ± 0.127
Handcrafted	Prototype	$\mathbf{0.268 \pm 0.041}$	0.448 ± 0.077	0.089 ± 0.017	0.170 ± 0.032	0.411 ± 0.072	0.009 ± 0.011
Handcrafted	Refined	0.232 ± 0.014	0.330 ± 0.033	0.134 ± 0.021	0.160 ± 0.017	0.343 ± 0.037	0.038 ± 0.025
Uniform Random Guessing		≈ 0.167 (6 classes)			≈ 0.200 (5 classes)		

1. **Target-exclusive activities scale with K:** Room A rises from 0.324 ± 0.078 at $K = 1$ to $\mathbf{0.510 \pm 0.053}$ at $K = 20$, Room B from 0.390 ± 0.093 to $\mathbf{0.588 \pm 0.068}$. Few-shot prototypes do capture novel activities.
2. **Shared transfer does not scale at all:** it sits near chance throughout (0.090 in Room A against chance 0.167; 0.211 in Room B against 0.200), so source-room prototypes never align with target-room signals however many target-exclusive labels are added.
3. **Unsupervised refinement hurts:** soft k-means lowers Room B Macro-F1 (0.337 → 0.295), because target embeddings cluster by session rather than activity and the updates drag prototypes off the true class centres.

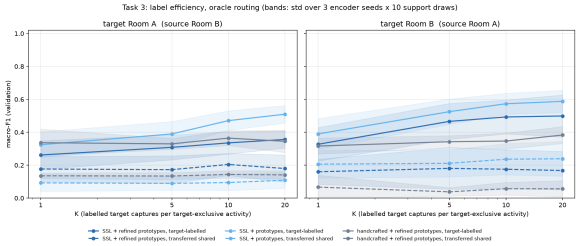


Figure 4: Task 3 label efficiency curves: Macro-F1 vs $K \in \{1, 5, 10, 20\}$ for Target Room A and Room B, showing scaling on target-exclusive classes vs stagnation on shared classes.

6.4 Optimal Hungarian Diagnostic & Confusion Matrix Analysis

To determine whether shared transfer failed due to poor prototype calibration or fundamental kinematic confusion, we computed the optimal bipartite Hungarian matching between predicted clusters and true validation labels (`src/task3_diagnostics.py`).

As shown in Table 5, even under an oracle permutation that optimally reassigns predicted classes to ground-truth labels, the upper bound on shared Macro-F1 reaches only **0.251** in Room A and **0.305** in Room B. In Room B, the optimal assignment swaps SIT and WALK (SIT ↔ WALK).

Inspection of the $K = 5$ confusion matrix (Figure 5) and the per-activity breakdown (Table 6) confirms this pathology: in Room B, **49.1**% of true SIT windows (368/750) and **37.5**% of STAND windows (281/750) are misclassified as WALK, while **35.3**% of WALK windows are classified as EMPTY. In Room A, static postures collapse into dynamic transitions: 34.3% of SIT windows are misclassified as SIT-DOWN, and 30.5% are misclassified as FALL. Thus, failure to transfer is not an algorithmic defect of the classifier, but a consequence of environment-specific multipath altering the frequency and variance characteristics of human motion.

Table 5: Hungarian optimal-assignment diagnostic: the best Macro-F1 reachable by any one-to-one relabelling of the predicted classes. The optimal permutation swaps SIT↔WALK in *both* directions (Room A additionally swaps STAND↔FALL); all other activities map to themselves. This is an *oracle* computed from validation labels — a diagnostic upper bound, never an achieved result.

Target Room	Macro-F1	Bound	Shared F1	Bound
Room A ($\mathcal{B} \to \mathcal{A}$)	0.243	0.255	0.177	0.251
Room B ($\mathcal{A} \to \mathcal{B}$)	0.292	0.366	0.182	0.305

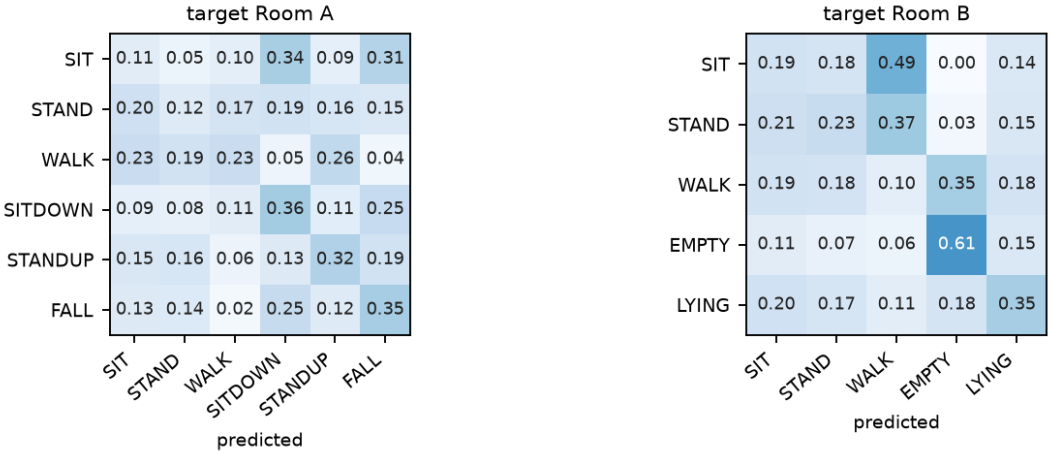


Figure 5: Task 3 row-normalised confusion at $K = 5$; configuration as in Table 6. Room B collapses SIT and STAND into WALK; Room A collapses static postures into transitions.

6.5 Routing Contribution Ablation

We evaluated the contribution of environment-aware routing by comparing three operational configurations: *oracle* routing sends each window to its true room's prototype bank; *predicted* routing uses the Task 1 classifier on the `task1_honest` preset (99.27% accuracy); *no* routing evaluates a single global bank over all eight activities without environment conditioning.

Predicted routing matches oracle routing to within **0.006** F1 (0.240 vs 0.240 in Room A; 0.343 vs 0.337 in Room B), while removing routing costs 6.0 points in Room A (0.240 → 0.180) and **14.2** in Room B (0.337 → 0.195), because target windows are then absorbed by source-exclusive prototypes that cannot apply (Room B windows

Table 6: Per-activity precision, recall and F1 at $K = 5$ (specification Section 11): SSL encoder, refined prototypes, oracle routing, pooled over 30 runs (3 seeds $\times$ 10 draws, 750 windows per class). Every shared activity — transferred with no target label — scores below every target-exclusive activity learned from just $K = 5$ captures.

Target Room A ($\mathcal{B} \to \mathcal{A}$)				**Target Room B** ($\mathcal{A} \to \mathcal{B}$)			
Activity	**P**	**R**	**F1**	**Activity**	**P**	**R**	**F1**
Shared — transferred from the source room, no target labels							
Sit	0.121	0.111	0.116	Sit	0.211	0.191	0.200
Stand	0.166	0.124	0.142	Stand	0.280	0.235	0.255
Walk	0.337	0.229	0.273	Walk	0.085	0.096	0.090
Target-exclusive — learned from $K = 5$ labelled captures							
Sit-Down	0.272	0.357	0.309	Empty	0.522	0.609	0.562
Stand-Up	0.300	0.320	0.309	Lying	0.360	0.348	0.354
Fall	0.274	0.349	0.307	—	—	—	—
Macro-F1			**0.243**	**Macro-F1**			**0.292**

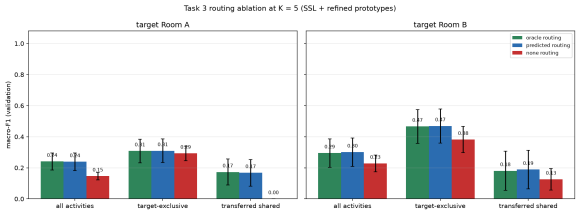


Figure 6: Routing ablation at $K = 5$: oracle vs predicted vs no routing, both target rooms.

predicted Fall). Routing protects the decision boundary — but since Task 1 succeeds on session artifacts, routing is close to free here and the oracle-versus-predicted gap is the only informative comparison.

7 Critical Limitations & Confounding Factors

In accordance with Section 12 of the challenge specification, we explicitly document six structural limitations of the benchmark:

1. **Room vs session (§12.1):** the two rooms were recorded in disjoint sessions, so any classifier also keys on oscillator drift, interference and transmission timing.
2. **Activity vs location (§12.2):** Room A activities were performed in fixed spatial sub-regions, so multipath fading correlates with the label.
3. **Two-room horizon (§12.3):** only $\mathcal{A} \leftrightarrow \mathcal{B}$ transfer is measured; generalisation to an unseen room is not established.
4. **Capture- not subject-disjoint (§12.4):** subjects overlap across splits, so subject-independent generalisation is untested.
5. **Not comparable to the original paper (§12.5):** its random 80/20 split leaks capture-level correlation; ours is capture-disjoint.
6. **Archive asymmetry (§12.6):** Experiment 2 is raw PCAP and Experiment 3 pre-extracted CSV, an upstream instrumentation difference we cannot remove.

8 Team Member Contributions & Logistics

The project was executed by Team Klique under a modular, test-driven engineering structure. Table 7 summarizes individual technical contributions.

Teaching Team Repository Access: Collaborator access to the GitHub repository has been granted to the course instructors: `sandipc-iitkgp`, `sayantan-kuila`, and `debjit2001`.

9 Mandatory AI Tool Use Disclosure

In compliance with the CS60055 academic integrity policy we disclose AI assistance in full, including cases where it was substantial rather than incidental.

Tools used. Anthropic Claude, driven interactively through Claude Code as an agentic pair-programming assistant with repository and shell access. All three members used this same assistant; no other AI tool was used on any part of this project.

Substantial assistance (Track 1: ingestion, normalisation, Task 1 and diagnostics). Working from the author's direction in extended interactive sessions, the assistant produced first-draft implementations of every module in the ingestion, quality-control, windowing, normalisation, Task 1 and diagnostics tracks (`pcap_reader`, `csi_decode`, `amplitude`, `csv_reader`, `cleaning`, `windowing`, `splits`, `normalization`, `dataset`, `config`, `task1_room`, `task1_ablation`, `task3_diagnostics`), the leakage test suite, `run_episode.py`,

Table 7: Team Member Contribution Statement.

Team Member	Core Technical Contributions & Responsibilities
R Mohan Poornachandra (23EX10027)	**Track 1 — ingestion, normalisation, Task 1:** clean-room PCAP parser and empirical BCM4366c0 bitfield resolution (`pcap_reader.py`, `csi_decode.py`); discovery of the Nexmon frame-interleaving artifact; deterministic windowing and manifest provenance (`windowing.py`); leakage-safe normalisation presets (`normalization.py`); Task 1 ablation ladder and session-confound analysis; the Room B WALK amplitude-shortcut diagnosis, the `per_window_scalar` scope behind `task2_level_free`, and the Hungarian assignment diagnostic (`task3_diagnostics.py`); technical report.
Hemant (22CS30029)	**Track 2 — representation learning:** 1D dilated residual CNN backbone (`encoder.py`); SimCLR/NT-Xent pretraining (`ssl_pretrain.py`); adjacency-breaking augmentations; 416-d handcrafted and random-weight baselines (`baselines.py`); frozen linear-probe cross-validation (`task2_probe.py`); UMAP diagnostics.
Saras Wagh (22CS30048)	**Track 3 — few-shot adaptation:** prototypical networks and soft k-means refinement (`fewshot.py`); capture-level support sampling and the K-sweep driver (`task3_adapt.py`); executed the full `task2_level_free` programme — encoder retraining and the complete Task 3 rerun on which all results in Sections 5–6 are based — and authored the retraction in Section 6.2; integrated the Hungarian diagnostic; routing ablation; confusion and label-efficiency figures.

`configs/experiment.yaml`, the repository documentation and Figure 1. Three findings reported in this paper also first surfaced during those sessions and were then reproduced and confirmed by the team: the empirical packed-CSI bitfield resolution (Section 2.2), the Nexmon frame-interleaving artifact and its consequence for the 150-observation window (Section 2.3), and the Room B WALK amplitude shortcut that forced the `task2_level_free` retraction (Section 6.2). The `per_window_scalar` normalisation scope introduced to remove that shortcut was likewise drafted with the assistant; the encoder retraining and the full Task 3 rerun on the resulting representation were carried out by the team.

Routine assistance. Debugging bitwise sign extension, pytest boilerplate, Matplotlib and LaTeX layout, and prose editing of this report, including verifying every reported number against the committed `results/*.csv` files.

Track 2 and Track 3 model code (`encoder.py`, `ssl_pretrain.py`, `baselines.py`, `task2_probe.py`, `fewshot.py`, `task3_adapt.py`) was written by the members credited in Table 7 using the same assistant in the same capacity: AI-drafted implementations followed by author review, execution and verification. The design decisions those files embody — the encoder topology, the NT-Xent objective and its temperature, the adjacency-breaking augmentation set, the handcrafted feature basis, the prototype and soft k-means formulations, the support-sampling protocol and the routing ablation — were made by those authors, who also ran every experiment reported from them.

Author responsibility. No AI-generated component was accepted without review. Every design decision, every rejected alternative and every claim in this report was checked by the responsible author against data the team regenerated; each member can explain the purpose and behaviour of every stage they are credited with, as Section 6 of the specification requires.

10 Conclusion

We engineered an independent, clean-room Python pipeline for cross-room WiFi activity recognition, and showed that both naive environment recognition and few-shot activity transfer are confounded by acquisition artifacts: Nexmon frame interleaving, software amplitude scaling and class-level loudness offsets. A 1D dilated residual CNN pretrained by SimCLR with adjacency-breaking augmentations reaches an 8-class Macro-F1 of $\mathbf{0.924 \pm 0.013}$, exceeding handcrafted kinematic features (0.893) and random weights (0.616). Prototypical networks learn novel target-exclusive activities from very few labels (F1 ≈ 0.59 at $K = 20$), but unlabelled shared-activity transfer fails through multipath-induced kinematic swaps. Cross-room WiFi sensing therefore requires active spatial-multipath calibration, not representation learning alone.

References

[1] IEEE DataPort, "CSI Human Activity Dataset," 2021. https://ieee-dataport.org/open-access/csi-human-activity.

[2] J. Schäfer, B. R. Barrsiwal, M. Kokhkharova, H. Adil, and J. Liebehenschel, "Human Activity Recognition Using CSI Information with Nexmon," *Applied Sciences*, vol. 11, no. 19, p. 8860, 2021.

[3] F. Gringoli *et al.*, "Free Your CSI: A CSI Extraction Platform for Modern Wi-Fi Chipsets," in *Proc. ACM WiNTECH*, 2019, pp. 21–28.

[4] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A Simple Framework for Contrastive Learning of Visual Representations," in *Proc. ICML*, 2020, pp. 1597–1607.

[5] J. Snell, K. Swersky, and R. Zemel, "Prototypical Networks for Few-shot Learning," in *Proc. NeurIPS*, vol. 30, 2017.

[6] H. Wang *et al.*, "Human Respiration Detection with Commodity WiFi Devices," in *Proc. ACM UbiComp*, 2015, pp. 25–36.

[7] J. Yang *et al.*, "Device-Free Human Activity Recognition Using Handheld Devices," *IEEE IoT J.*, vol. 5, no. 2, pp. 796–805, 2018.

[8] H. Li *et al.*, "Wi-Motion: Robust Human Activity Recognition Using WiFi Signals," *IEEE IoT J.*, vol. 6, no. 4, pp. 6934–6946, 2019.

[9] F. Pedregosa *et al.*, "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[10] A. Paszke *et al.*, "PyTorch: An Imperative Style, High-Performance Deep Learning Library," in *Proc. NeurIPS*, vol. 32, 2019.